

Backup and Recovery



Module 4.2

Backup and Recovery

Upon completion of this module, you will be able to:

- Describe best practices for planning Backup and Recovery.
- Describe the common media and types of data that are part of a Backup and Recovery strategy.
- Describe the common Backup and Recovery topologies.
- Describe the Backup and Recovery Process.
- Describe Management considerations for Backup and Recovery.



In this module ...

This module contains the following lessons:

- Planning for Backup and Recovery
- Backup and Recovery Methods
- Backup Architecture Topologies
- Managing the Backup Process



Lesson: Planning for Backup and Recovery

Upon completion of this lesson, you be able to:

- Define Backup and Recovery.
- Describe common reasons for a Backup and Recovery plan.
- Describe the business considerations for Backup and Recovery.
- Define RPO and RTO.
- Describe the data considerations for Backup and Recovery
- Describe the planning for Backup and Recovery.



What is a Backup?

- Backup is an additional copy of data that can be used for restore and recovery purposes.
- The Backup copy is used when the primary copy is lost or corrupted.
- This Backup copy can be created as a:
 - Simple copy (there can be one or more copies)
 - Mirrored copy (the copy is always updated with whatever is written to the primary copy.)

Backup and Recovery Strategies

Several choices are available to get the data to the backup media such as:

- Copy the data.
- Mirror (or snapshot) then copy.
- Remote backup.
- Copy then duplicate or remote copy.

It's All About Recovery!

- Businesses back up their data to enable its recovery in case of potential loss.
- Businesses also back up their data to comply with regulatory requirements.
- Types of backup derivatives:
 - Disaster Recovery
 - Archival
 - Operational

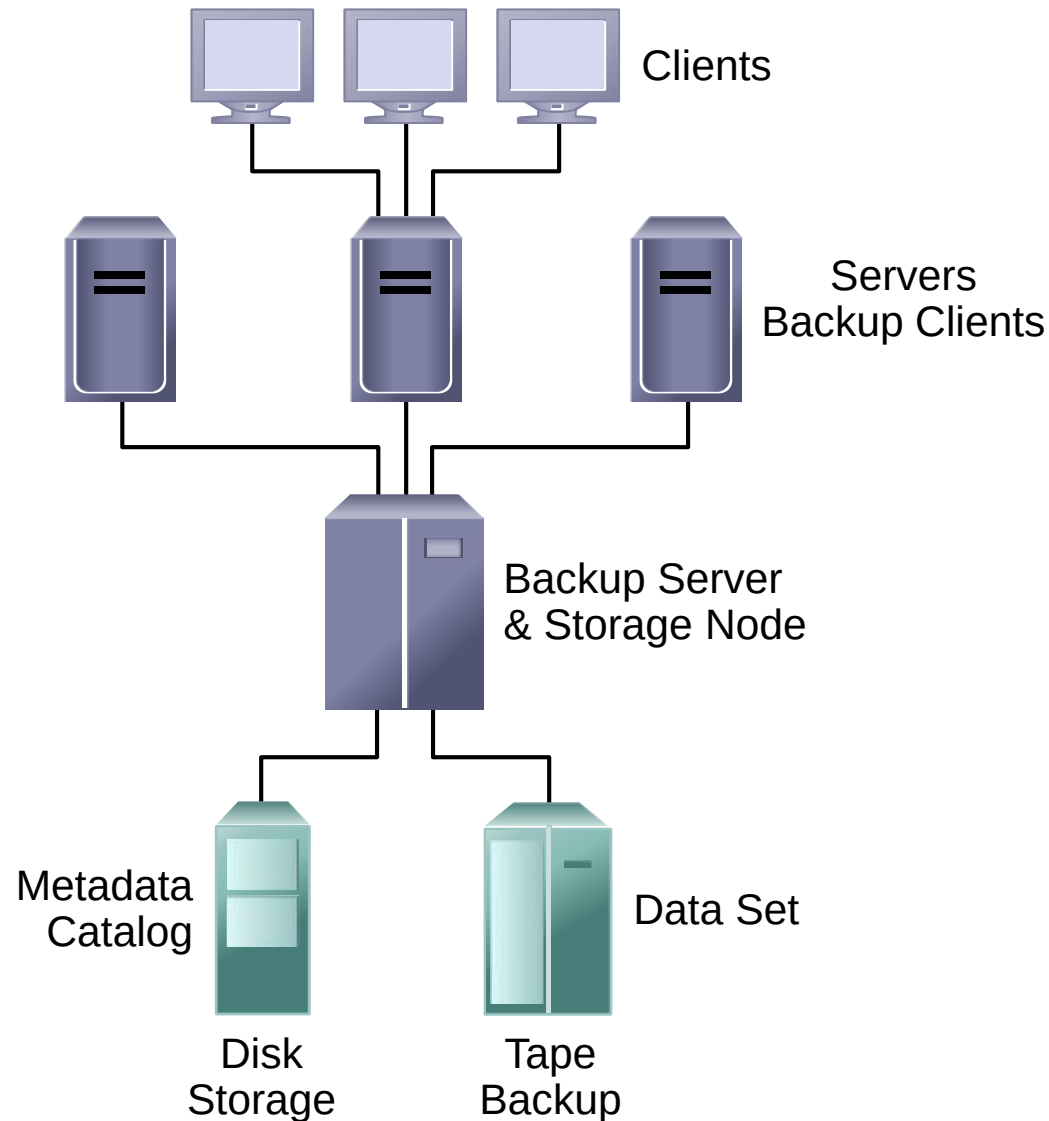
Reasons for a Backup Plan

- Hardware Failures
- Human Factors
- Application Failures
- Security Breaches
- Disasters
- Regulatory and Business Requirements

How does Backup Work?

- Client/Server Relationship
- Server
 - Directs Operation
 - Maintains the Backup Catalog
- Client
 - Gathers Data for Backup (a backup client sends backup data to a backup server or storage node).
- Storage Node

How does Backup Work, continued



Business Considerations

- Customer business needs determine:
 - What are the restore requirements – RPO & RTO?
 - Where and when will the restores occur?
 - What are the most frequent restore requests?
 - Which data needs to be backed up?
 - How frequently should data be backed up?
 - hourly, daily, weekly, monthly
 - How long will it take to backup?
 - How many copies to create?
 - How long to retain backup copies?

Data Considerations: File Characteristics

- Location
- Size
- Number

Data Considerations: Data Compression

Compressibility depends on the data type, for example:

- Application binaries – do not compress well.
- Text – compresses well.
- JPEG/ZIP files – are already compressed and expand if compressed again.

Data Considerations: Retention Periods

- Operational
 - Data sets on primary media (disk) up to the point where most restore requests are satisfied, then moved to secondary storage (tape).
- Disaster Recovery
 - Driven by the organization's disaster recovery policy
 - Portable media (tapes) sent to an offsite location / vault.
 - Replicated over to an offsite location (disk).
 - Backed up directly to the offsite location (disk, tape or emulated tape).
- Archiving
 - Driven by the organization's policy.
 - Dictated by regulatory requirements.

Lesson: Summary

Topics in this lesson included:

- Backup and Recovery definitions and examples.
- Common reasons for Backup and Recovery.
- The business considerations for Backup and Recovery.
- Recovery Point Objectives and Recovery Time Objectives.
- The data considerations for Backup and Recovery
- The planning for Backup and Recovery.



Lesson: Backup and Recovery Methods

Upon completion of this lesson, you be able to:

- Describe Hot and Cold Backups.
- Describe the levels of Backup Granularity.

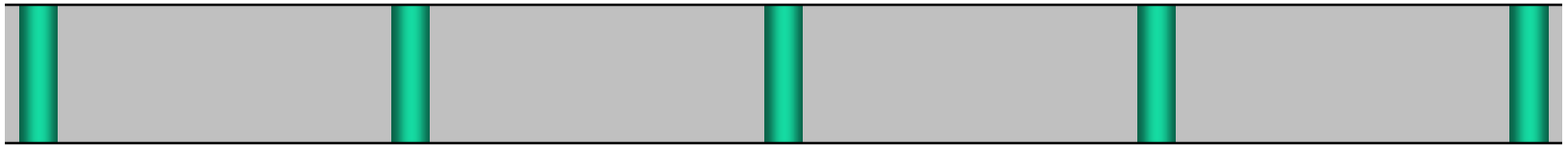


Database Backup Methods

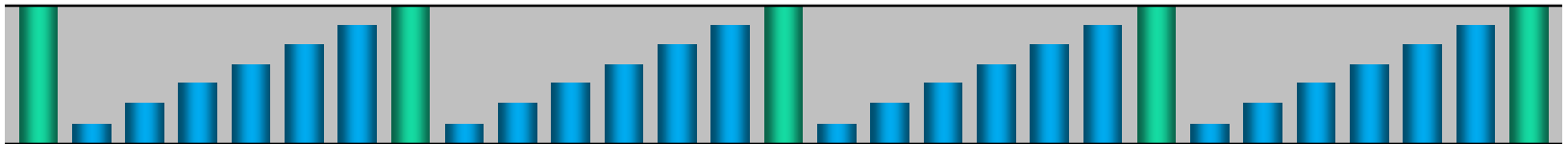
- Hot Backup: production **is not** interrupted.
- Cold Backup: production **is** interrupted.
- Backup Agents manage the backup of different data types such as:
 - Structured (such as databases)
 - Semi-structured (such as email)
 - Unstructured (file systems)

Backup Granularity and Levels

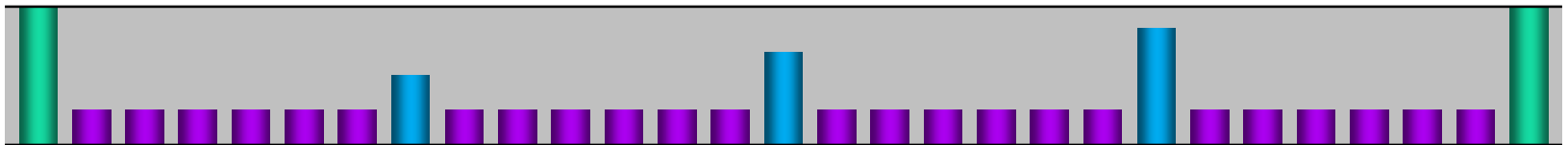
Full Backup



Cumulative (Differential)



Incremental



Full

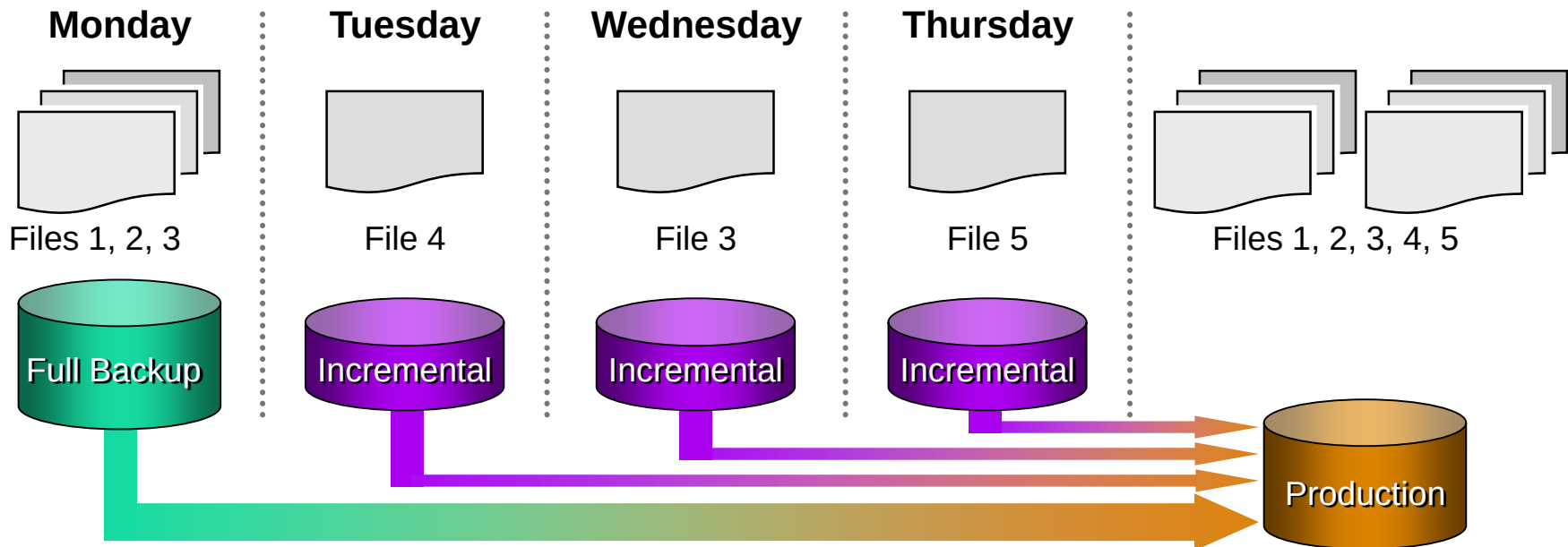


Cumulative



Incremental

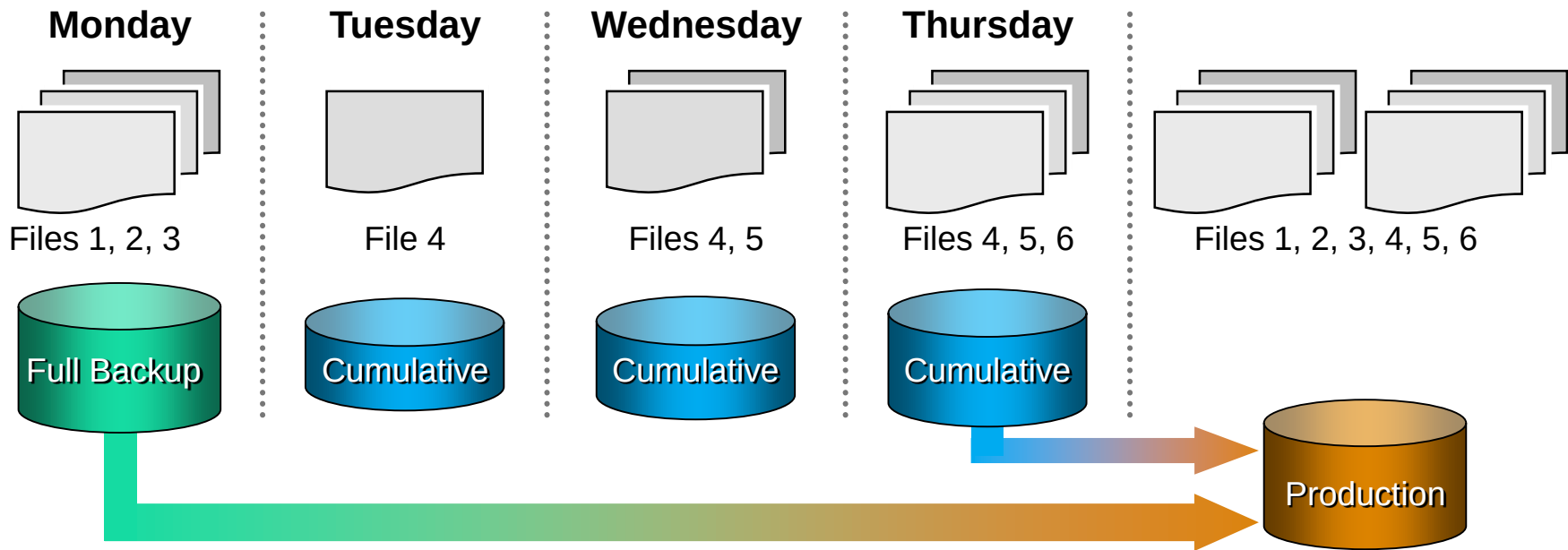
Restoring an Incremental Backup



• Key Features

- Files that have changed since the last full or incremental backup are backed up.
- Fewest amount of files to be backed up, therefore faster backup and less storage space.
- Longer restore because last full and all subsequent incremental backups must be applied.

Restoring a Cumulative Backup



- **Key Features**

- More files to be backed up, therefore it takes more time to backup and uses more storage space.
- Much faster restore because only the last full and the last cumulative backup must be applied.

Lesson: Summary

Topics in this lesson included:

- Hot and Cold Backups.
- The levels of Backup Granularity.



Lesson: Backup Architecture Topologies

Upon completion of this lesson, you be able to:

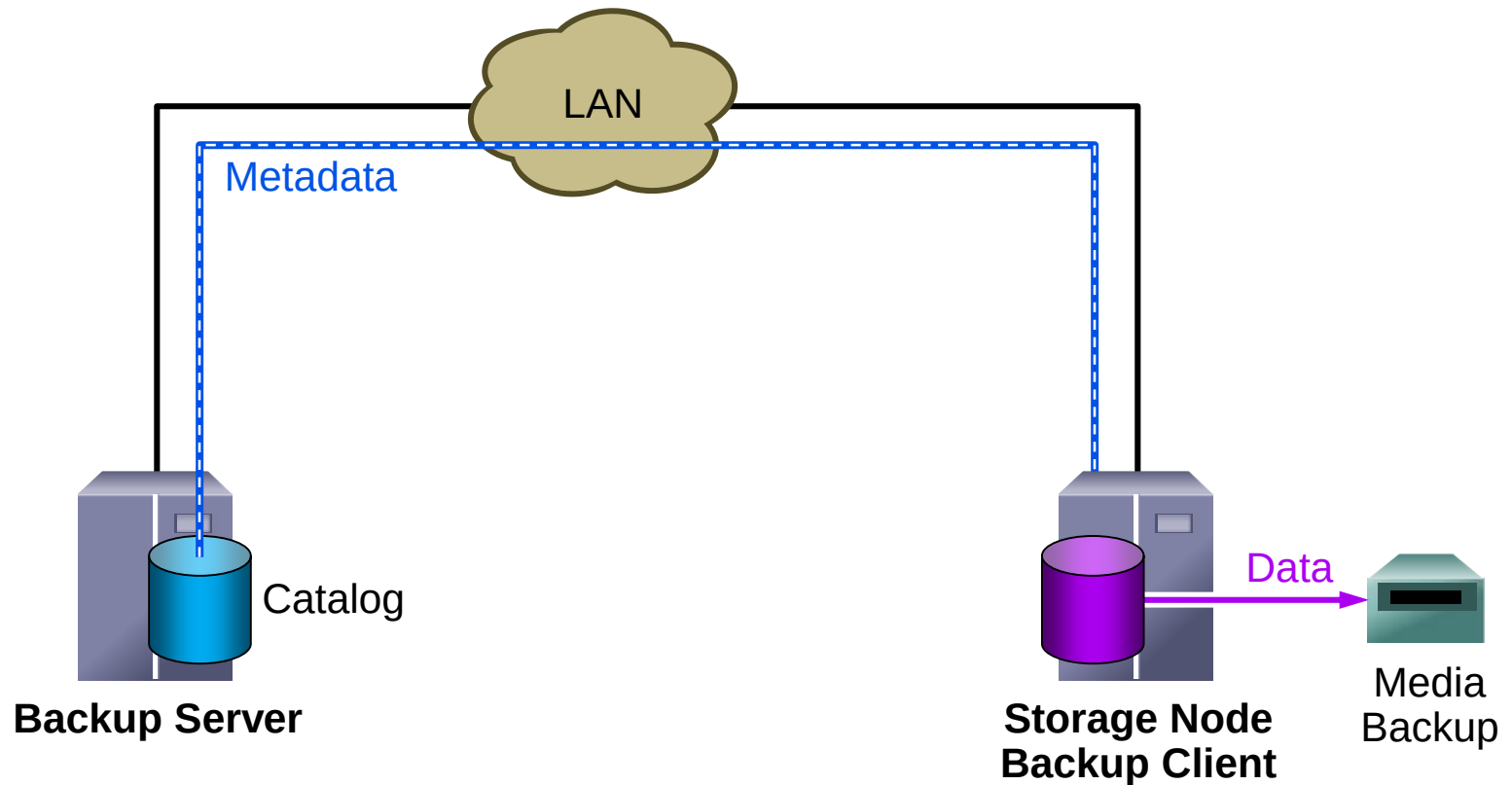
- Describe DAS, LAN, SAN, Mixed topologies.
- Describe backup media considerations.



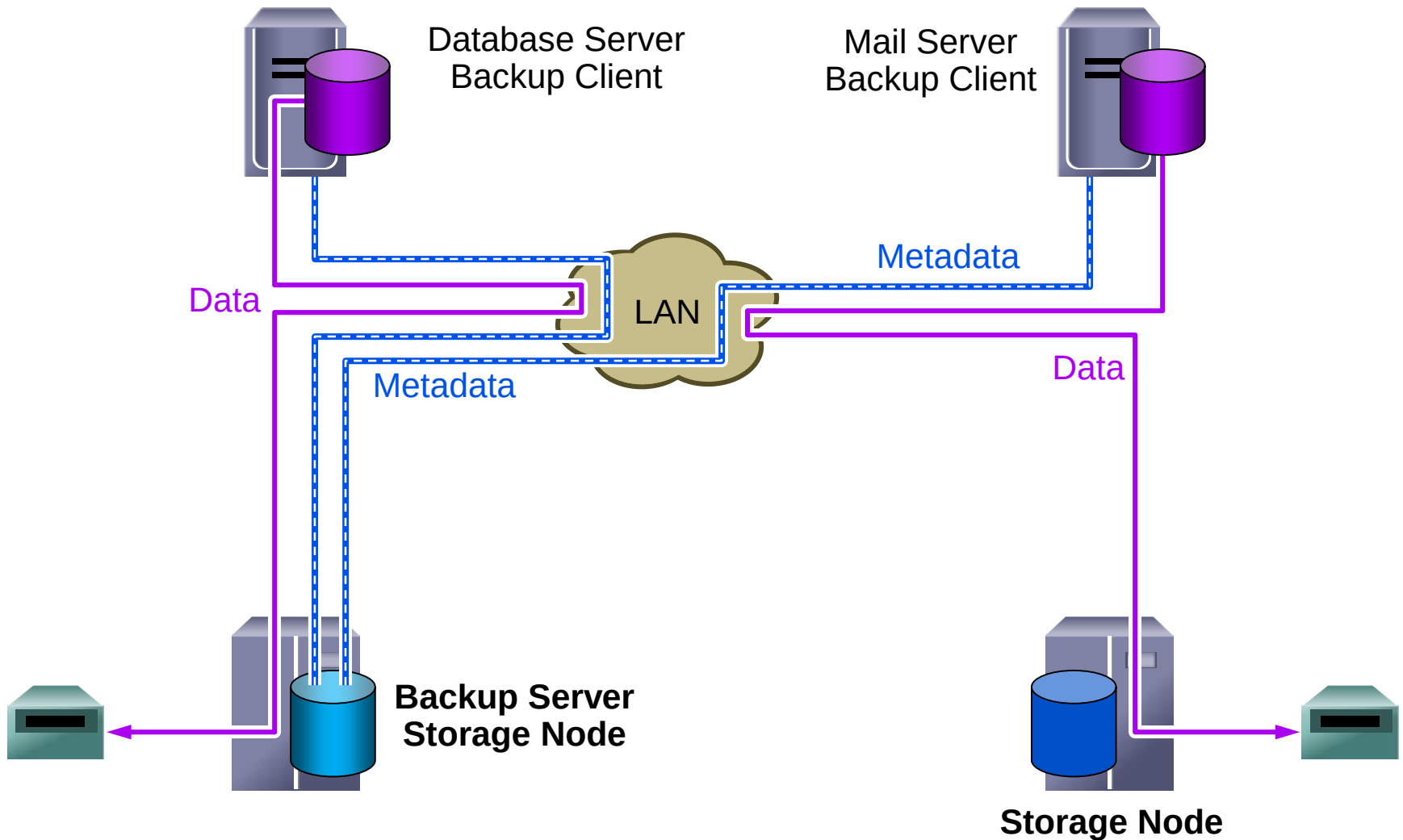
Backup Architecture Topologies

- There are 3 basic backup topologies:
 - Direct Attached Based Backup
 - LAN Based Backup
 - SAN Based Backup
- These topologies can be integrated, forming a “mixed” topology

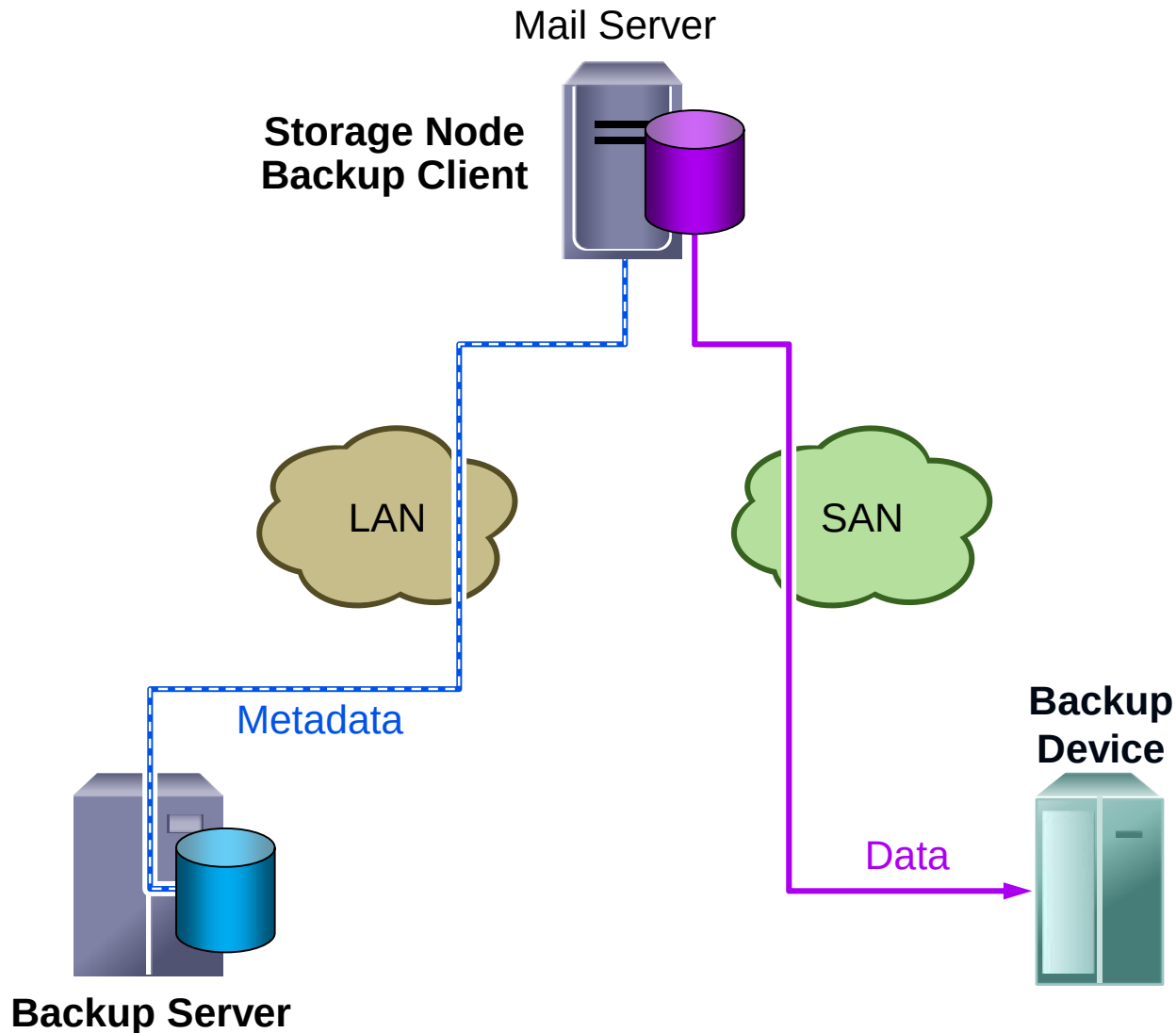
Direct Attached Based Backups



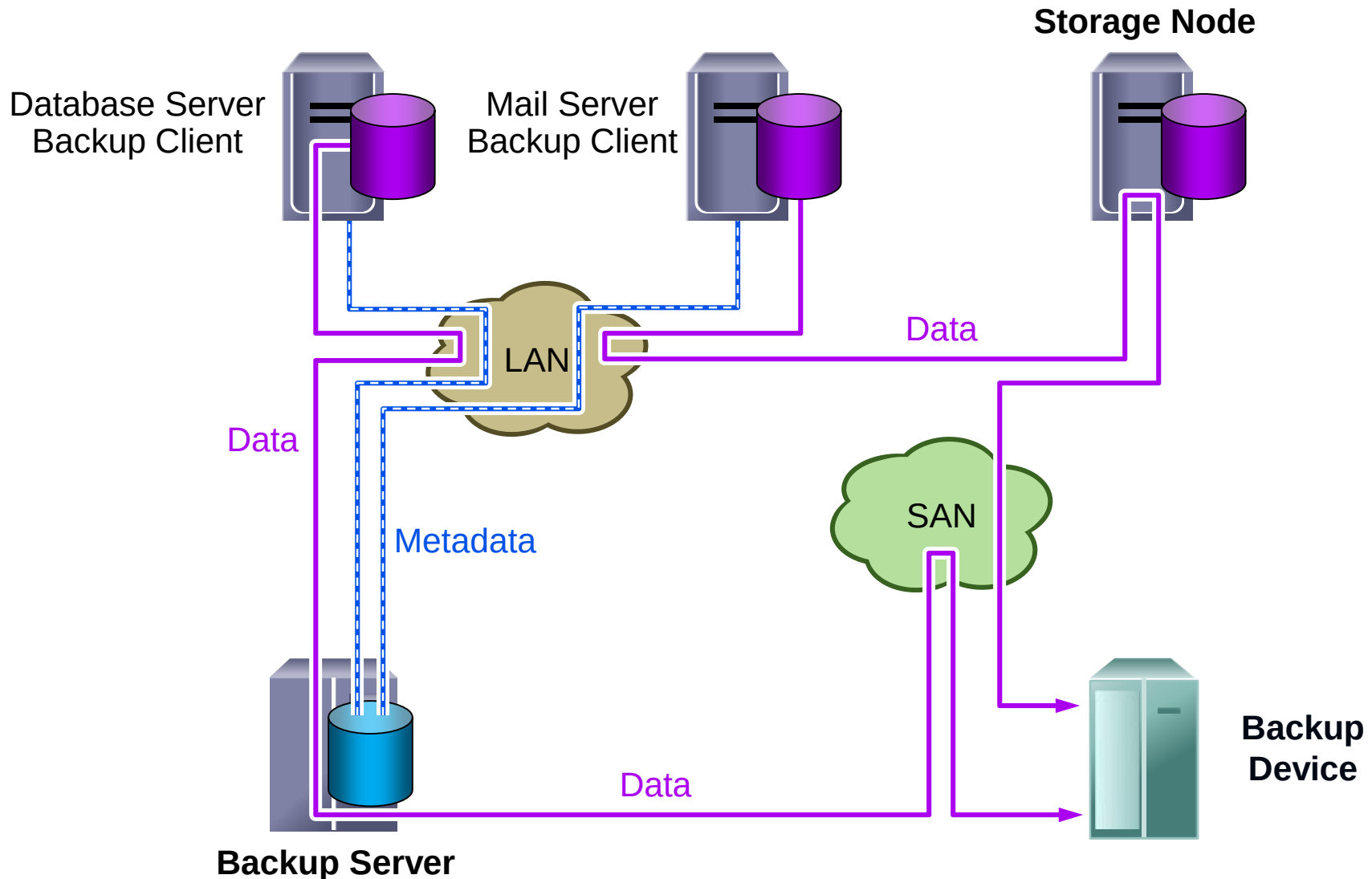
LAN Based Backups



SAN Based Backups (LAN Free)



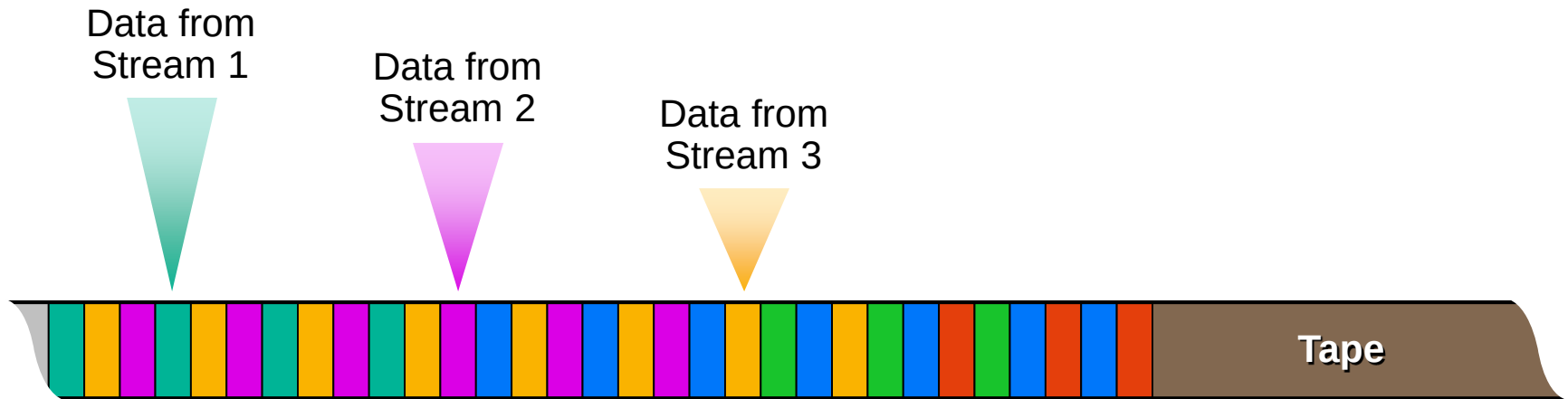
SAN/LAN Mixed Based Backups



Backup Media

- Tape
 - Traditional destination for backups
 - Sequential access
 - No protection
- Disk
 - Random access
 - Protected by the storage array (RAID, hot spare, etc)

Multiple Streams on Tape Media

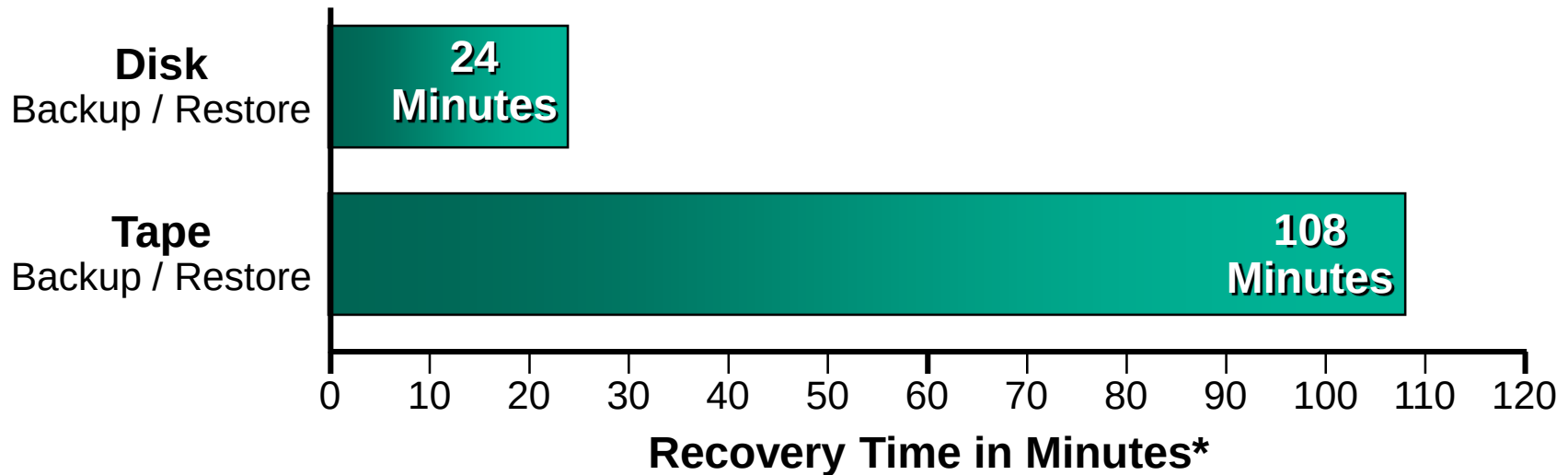


- Multiple streams interleaved to achieve higher throughput on tape
 - Keeps the tape streaming, for maximum write performance
 - Helps prevent tape mechanical failure
 - Greatly increases time to restore

Backup to Disk

- Backup to disk *minimizes* tape in backup environments by using disk as the primary *destination* device
 - Cost benefits
 - No processes changes needed
 - Better service levels
- Backup to disk aligns backup strategy to RTO and RPO

Tape versus Disk – Restore Comparison



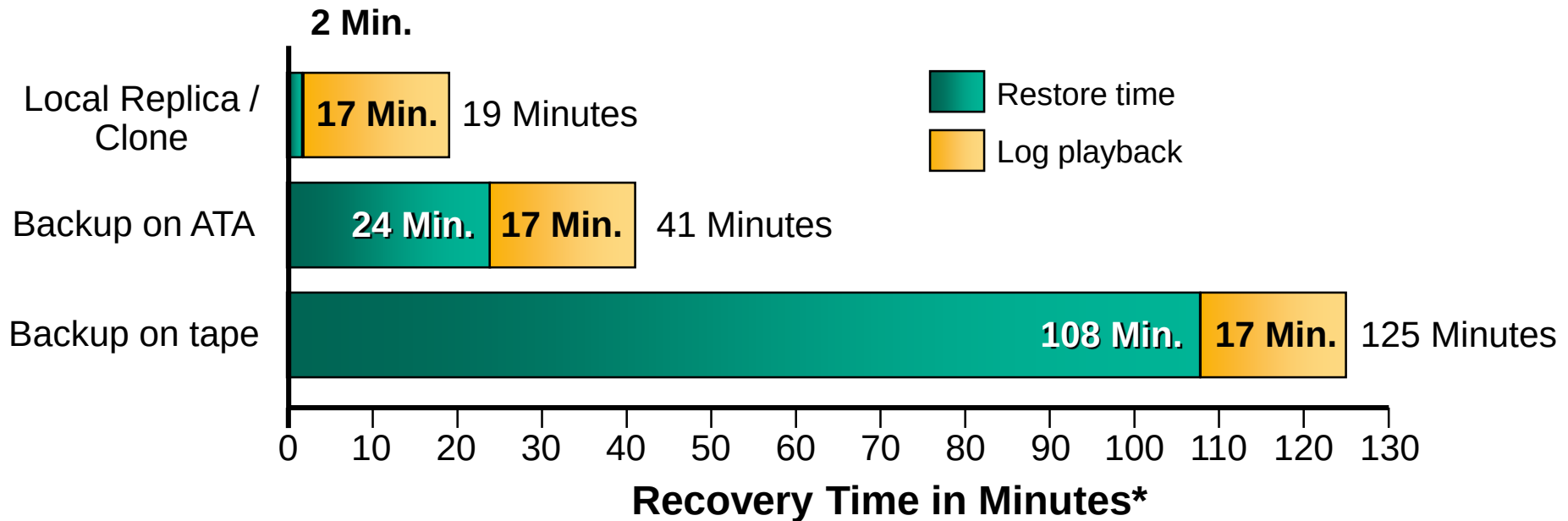
*Total time from point of failure to return of service to e-mail users

Typical Scenario:

- 800 users, 75 MB mailbox
- 60 GB database

Source: EMC Engineering and EMC IT

Three Backup / Restore Solutions based on RTO

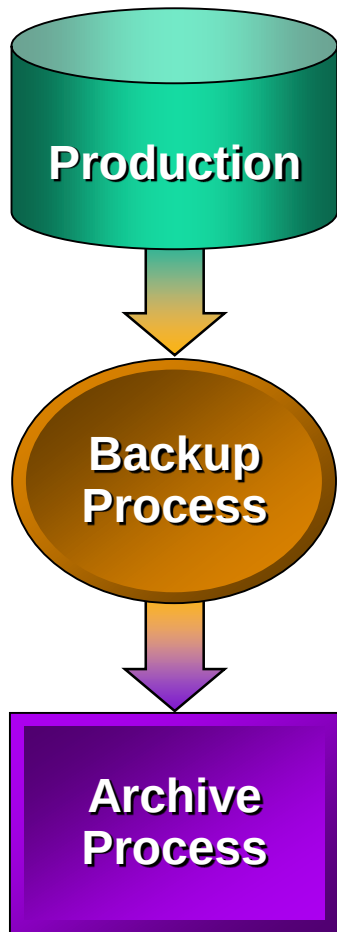


*Total time from point of failure to return of service to e-mail users

Typical Scenario:

- 800 users, 75 MB mailbox
 - 60 GB DB – restore time
 - 500 MB logs – log playback
- Time of last image dictates the log playback time
 - Larger data sets extend the recovery time (ATA and tape)

Traditional Backup, Recovery and Archive Approach



- Production environment grows
 - Requires constant tuning and data placement to maintain performance
 - Need to add more tier-1 storage
- Backup environment grows
 - Backup windows get longer and jobs do not complete
 - Restores take longer
 - Requires more tape drives and silos to keep up with service levels
- Archive environment grows
 - Impact flexibility to retrieve content when requested
 - Requires more media, adding management cost
 - No investment protection for long term retention requirements

Differences Between Backup / Recovery & Archive

Backup / Recovery

A **secondary copy** of information

Used for **recovery** operations

Improves availability by enabling application to be restored to a specific point in time

Typically **short-term** (weeks or months)

Data typically **overwritten** on periodic basis (e.g., monthly)

Not for regulatory compliance—though some are forced to use

Archive

Primary copy of information

Available for information **retrieval**

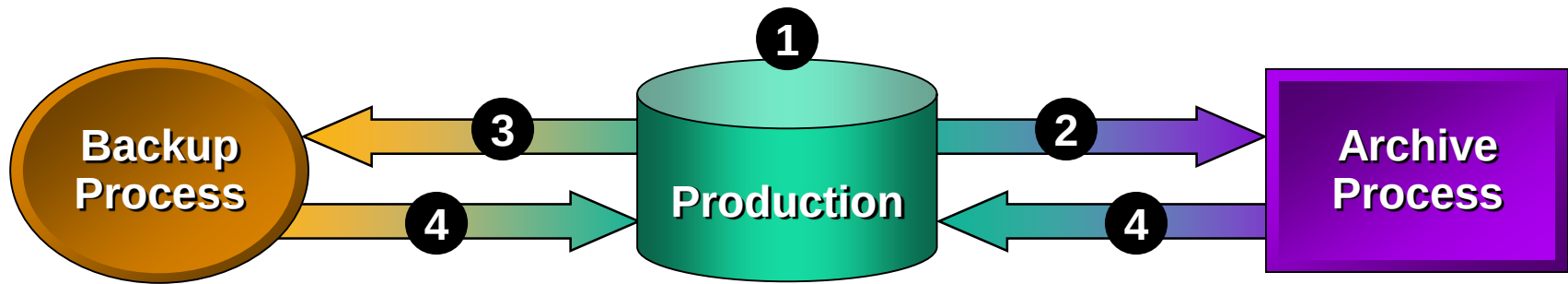
Adds operational efficiencies by moving fixed / unstructured content out of operational environment

Typically **long-term** (months, years, or decades)

Data typically **maintained** for analysis, value generation, or compliance

Useful for compliance and should take into account information-retention policy

New Architecture for Backup, Recovery & Archive



- ❶ Understand the environment
- ❷ Actively archive valuable information to tiered storage
- ❸ Back up active production information to disk
- ❹ Retrieve from archive or recover from backup

Lesson: Summary

Topics in this lesson included:

- The DAS, LAN, SAN, and Mixed topologies.
- Backup media considerations.



Lesson: Managing the Backup Process

Upon completion of this lesson, you be able to:

- Describe features and functions of common Backup/Recovery applications.
- Describe the Backup/Recovery process management considerations.
- Describe the importance of the information found in Backup Reports and in the Backup Catalog.



How a Typical Backup Application Works

- Backup clients are grouped and associated with a Backup schedule that determines when and which backup type will occur.
- Groups are associated with Pools, which determine which backup media will be used.
- Each backup media has a unique label.
- Information about the backup is written to the Backup Catalog during and after it completes. The Catalog shows:
 - when the Backup was performed, and
 - which media was used (label).
- Errors and other information is also written to a log.

Backup Application User Interfaces

There are typically two types of user interfaces:

- Command Line Interface – CLI
- Graphical User Interfaces – GUI

Managing the Backup and Restore Process

- Running the B/R Application: Backup
 - The backup administrator configures it to be started, most (if not all) of the times, automatically
 - Most backup products offer the ability for the backup client to initiate their own backup (usually disabled)
- Running the B/R Application: Restore
 - There is usually a separate GUI to manage the restore process
 - Information is pulled from the backup catalog when the user is selecting the files to be restored
 - Once the selection is finished, the backup server starts reading from the required backup media, and the files are sent to the backup client

Backup Reports

- Backup products also offer reporting features.
- These features rely on the backup catalog and log files.
- Reports are meant to be easy to read and provide important information such as:
 - Amount of data backed up
 - Number of completed backups
 - Number of incomplete backups (failed)
 - Types of errors that may have occurred
- Additional reports may be available, depending on the backup software product used.

Importance of the Backup Catalog

- As you can see, backup operations strongly rely on the backup catalog
- If the catalog is lost, the backup software alone has no means to determine where to find a specific file backed up two months ago, for example
- It can be reconstructed, but this usually means that all of the backup media (i.e. tapes) have to be read
- It's a good practice to protect the catalog
 - By replicating the file system where it resides to a remote location
 - By backing it up
- Some backup products have built-in mechanisms to protect their catalog (such as automatic backup)

Lesson: Summary

Topics in this lesson included:

- The features and functions of common Backup/Recovery applications.
- The Backup/Recovery process management considerations.
- The importance of the information found in Backup Reports and in the Backup Catalog.



Module Summary

Key points covered in this module:

- The best practices for planning Backup and Recovery.
- The common media and types of data that are part of a Backup and Recovery strategy.
- The common Backup and Recovery topologies.
- The Backup and Recovery Process.
- Management considerations for Backup and Recovery.



✓ *Check Your Knowledge*

- What are three reasons for doing a Backup plan?
- What are the three topologies that support creating a Backup set?
- What are the advantages and disadvantages of using tape as the Backup media?
- What are the three levels of granularity found in Backups?

Apply Your Knowledge...

Upon completion of this topic, you will be able to:

- Describe EMC's product implementation of a Backup and Recovery solution.

EMC NetWorker

Tiered Protection and Recovery Management

Remove risk

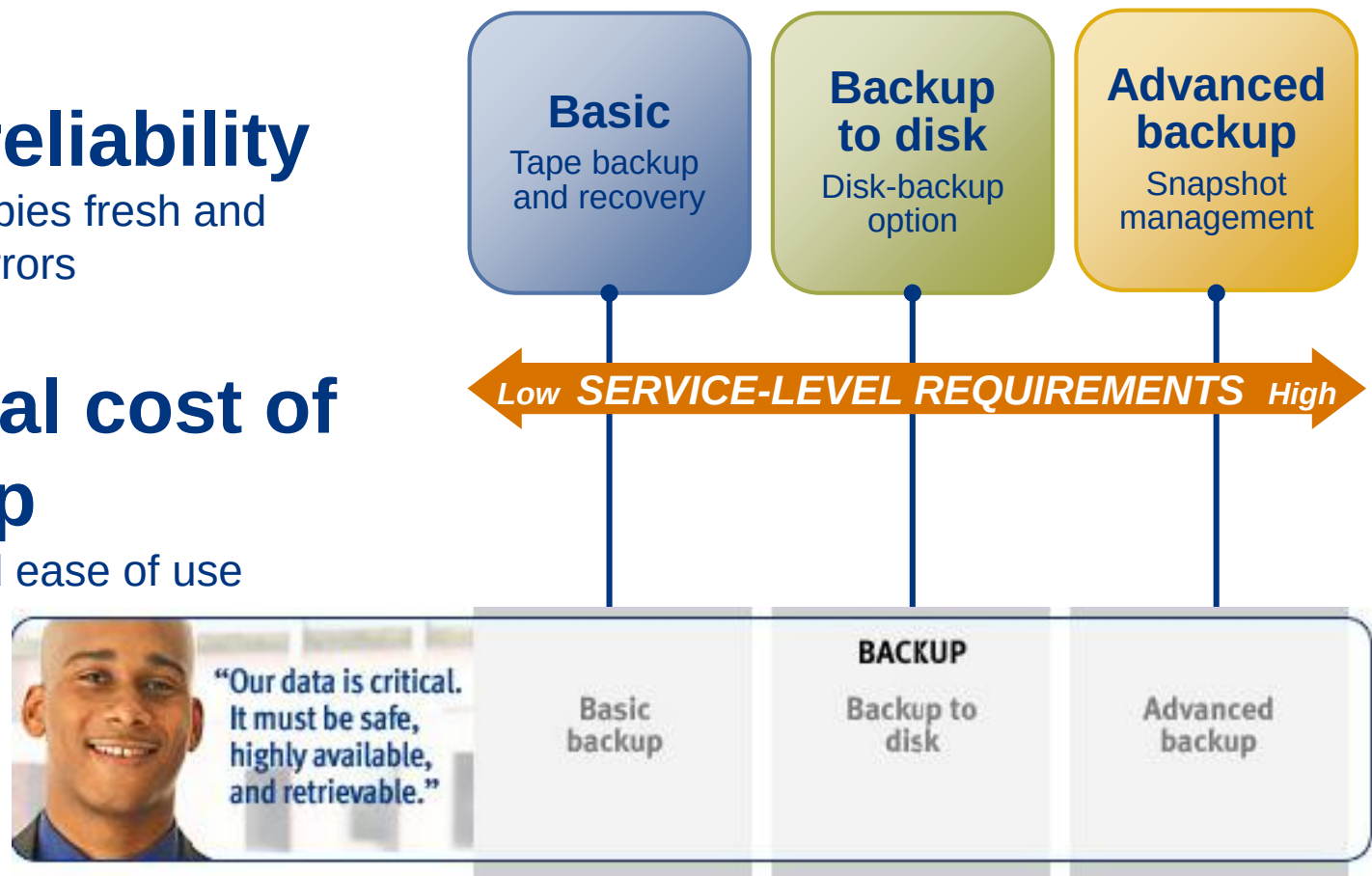
Faster and more consistent data backup

Improve reliability

Keep recovery copies fresh and reduce process errors

Lower total cost of ownership

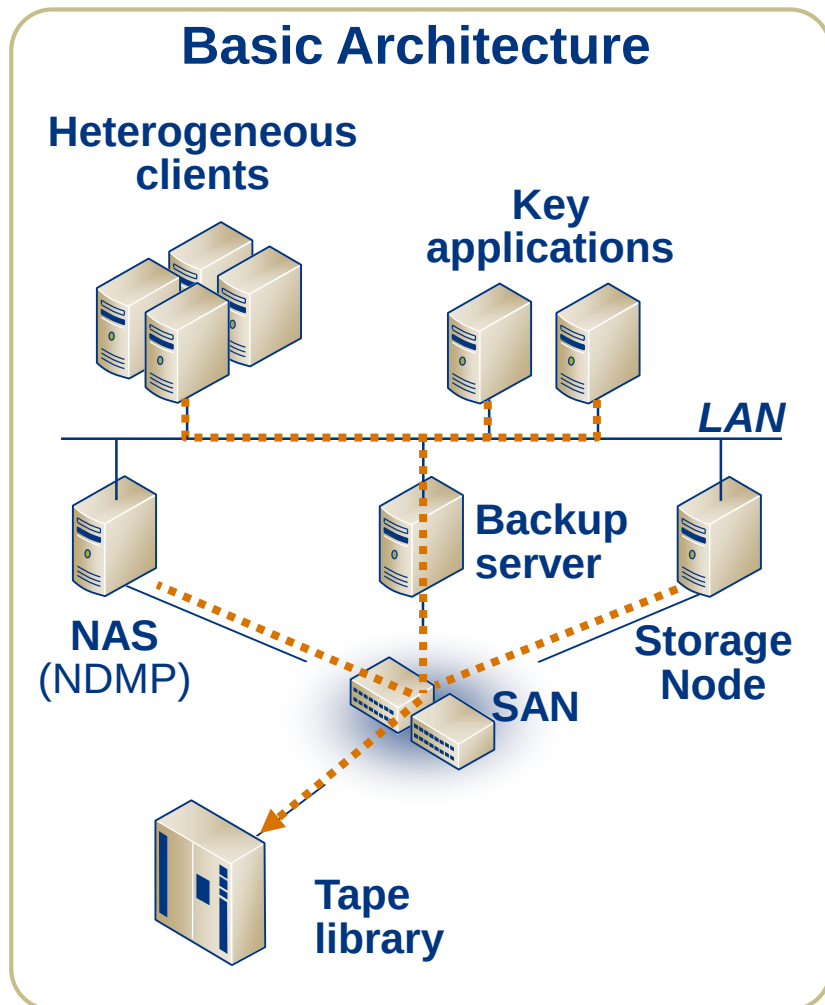
Centralization and ease of use



NetWorker Backup and Recovery

Solution Features

- Enterprise protection
 - Critical applications
 - Heterogeneous platforms and storage
 - Scalable architecture
 - 256-bit AES encryption and secure authentication
- Centralized management
 - Graphical user interface
 - Customizable reporting
 - Wizard-driven configuration
- Performance
 - Data multiplexing
 - Advanced indexing
 - Efficient media management



Critical Application and Database Protection

Backup without Application Modules

Offline (Cold)

Shut down application



Back up application



Restart application



Backup with NetWorker Application Modules



NetWorker MODULE

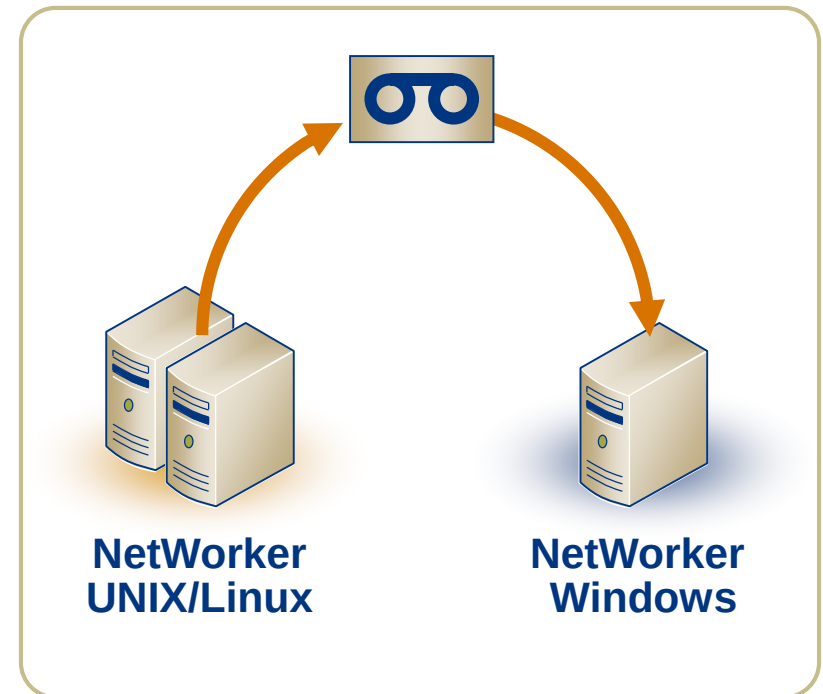
24x7 OPERATIONS



Integration with application APIs for backup and recovery

Media-Management Advantages

- Open Tape Format
 - Datastream multiplexing
 - Self-contained indexing
 - Cross-platform format
 - UNIX ↔ Windows ↔ Linux
 - Minimize impact of tape corruption
- Dynamic drive sharing
 - Cross-platform tape-drive sharing
 - On-demand device usage
 - Reduce hardware total cost of ownership

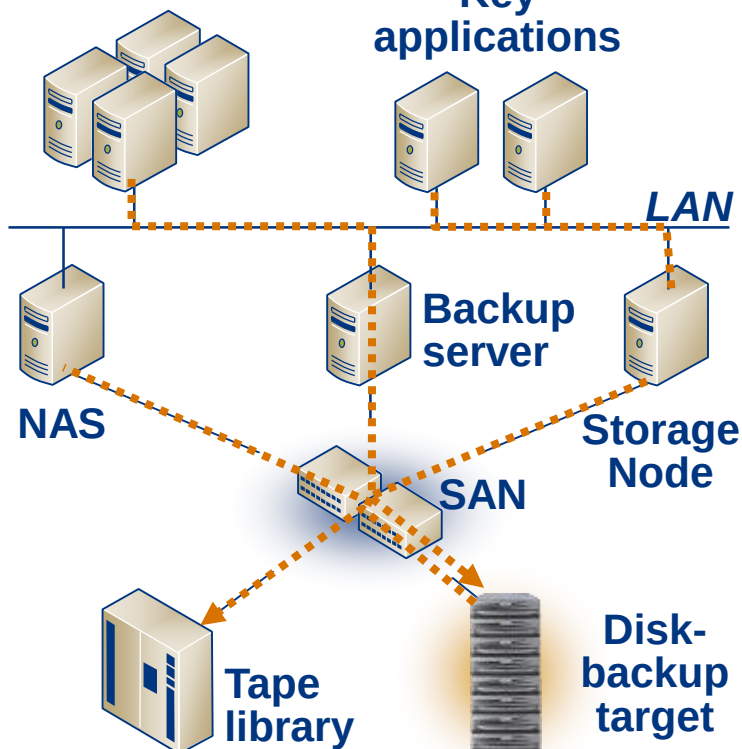


NetWorker DiskBackup Option

Backup-to-Disk Architecture

Heterogeneous clients

Key applications

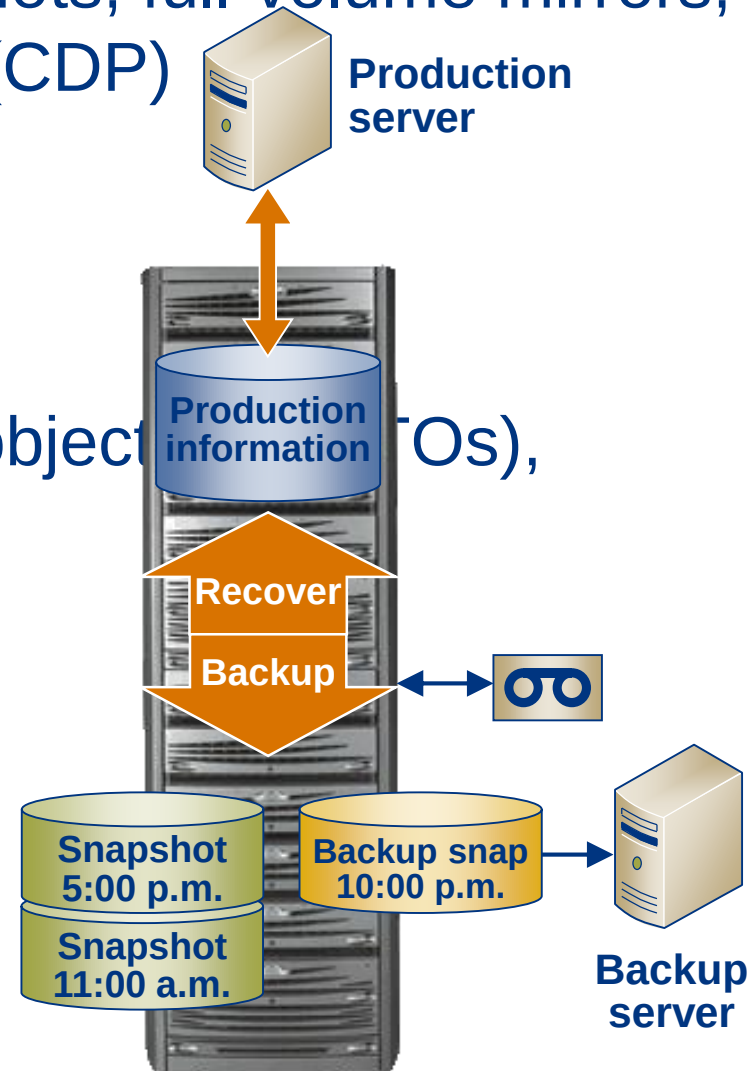


- High performance
 - Simultaneous-access operations
 - No penalty on restore versus tape
- Policy-based migration of data from disk to tape
 - Automated staging and cloning
 - Up to 50% faster
 - Clone backups jobs as they complete
 - Reduce wear and tear on tape drives and cartridges
- Superior capability
 - Operational backup and recovery for all clients, including NAS with NDMP
 - Direct file access for fast recovery

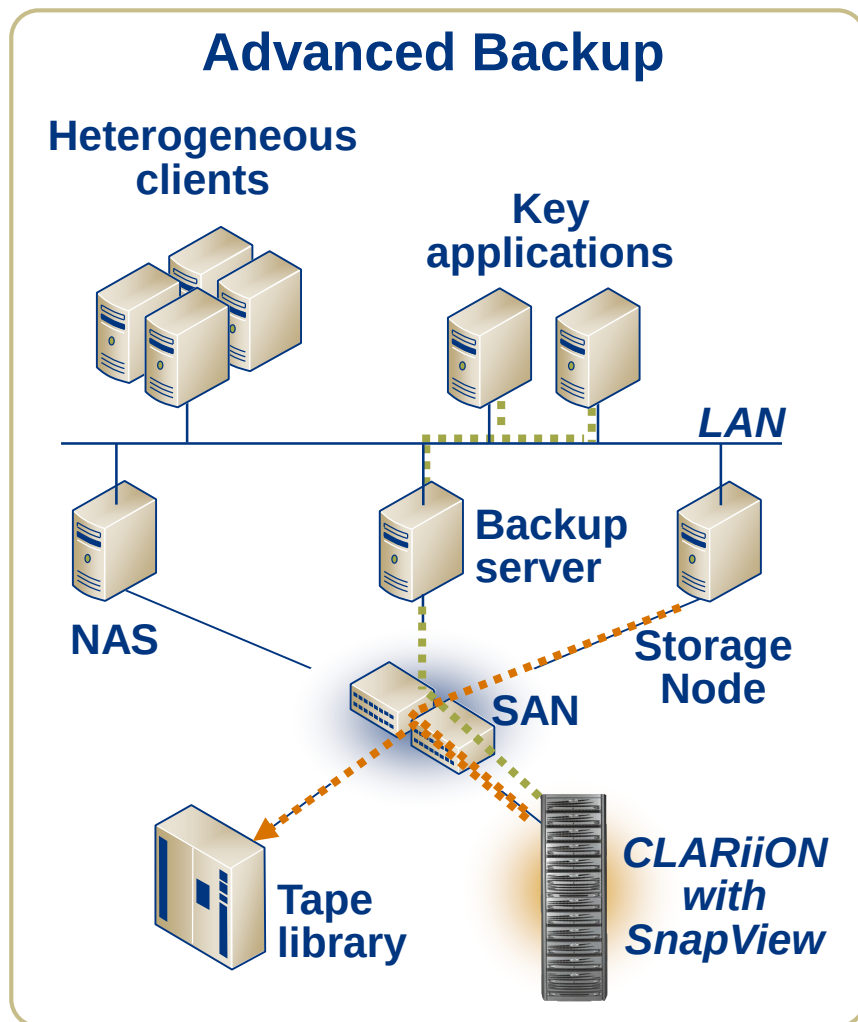
Advanced Backup - Snapshots and CDP

- Integration of backup with snapshots, full-volume mirrors, and Continuous Data Protection (CDP)
- Instant restore
- Off-host backups
- Achieve stringent recovery-time objectives (RTOs), recovery-point objectives (RPOs)

It is expected that *snapshot technology* for data protection will surpass backup to tape as the trend in data protection as organizations continue to focus on *recovery times*



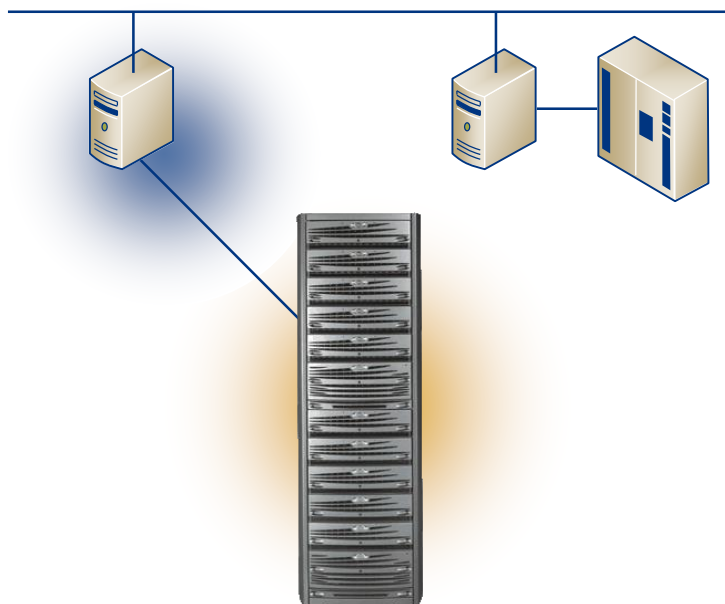
NetWorker PowerSnap Module



- Policy-based management
 - Administer snapshots in NetWorker
 - Schedule, create, retain, and delete snapshots by policy
- Third-party integration
 - Leverage third-party replication technology
 - Array-based (Symmetrix DMX, CLARiiON CX, etc.)
 - Software-based (RecoverPoint)
- Application recovery
 - Integration with Application Modules to ensure consistent state
 - Exchange / SQL / Oracle / SAP

NetWorker SnapImage Module

Advanced Backup



10,000,000+ files
1,000,000+ directories

- Block-level backups
 - Host-based snapshot
 - Targeted at high-density file systems
 - Single-file restore
 - Sparse backups
- High performance
 - Significant backup-and-restore performance impact—up to 10 times faster
 - Drive tape at rated speeds
 - Optional network-accelerated serverless backup with Cisco intelligent switch

Solution Example: Major Telecom Company

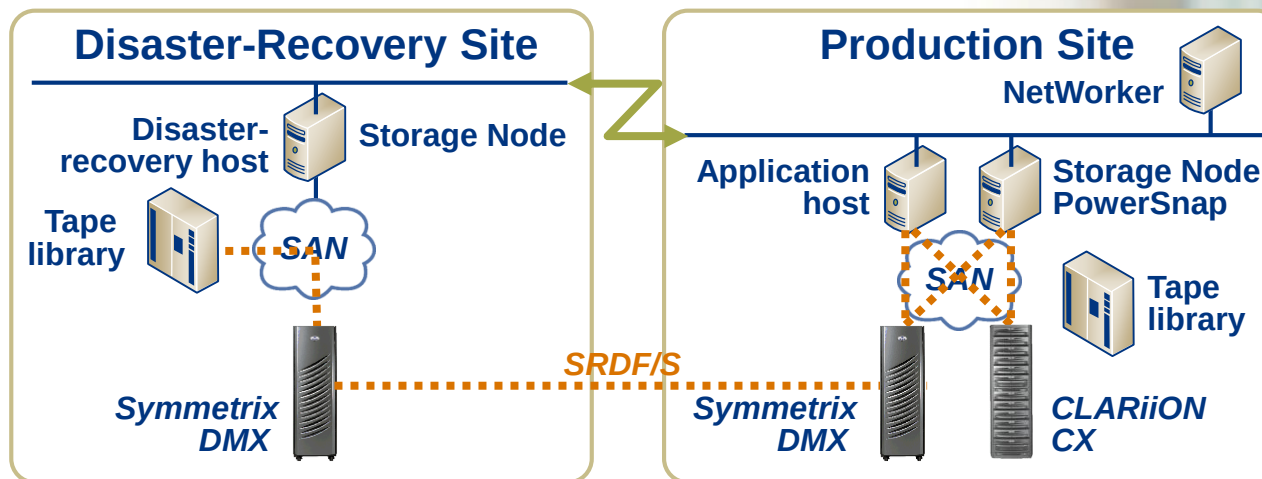
Enterprise-Information Protection

Business Challenge:

- Complex application environment
- No backup window
- Recovery-time objective: Restore 24 TB in two hours

Solution:

- NetWorker PowerSnap with Symmetrix and TimeFinder/Snap
 - Server-free backup
- NetWorker DiskBackup Option with CLARiiON with ATA disks
 - Rapid primary-site protection
- NetWorker and SRDF/S
 - Disaster recovery
 - Offsite protection



Value Proposition

Zero backup window for applications
Eliminated data-loss risk
Reduced management overhead